Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	We Need New Laptops
Project / Solution Name	JobTransit
Name (Leader)	Evelyn Ang
Email & Phone (Leader)	evelynang937@gmail.com , 0167606839
University Affiliation(s)	Universiti Teknologi Malaysia
Live Prototype URL	https://job-transit.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/yaotingchun/JobTransit

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

JobTransit operates on a decoupled modern cloud architecture designed for low latency and privacy:

1. Frontend / User Interface: Built with React 19, TypeScript, Vite, and Tailwind CSS. The client performs in-browser document processing using PDF.js and Tesseract.js (OCR), extracting raw text directly on the user's machine to reduce cloud bandwidth and protect sensitive documents.
2. AI Layer (GenAI): The structured text is transmitted via HTTPS to the Google Gemini API (gemini-2.0-flash / gemini-2.5-flash) using constrained JSON schemas. Gemini extracts verified technical skills, determines competency fits, and generates dynamic clarification questions for ambiguous abilities.
3. Backend Services: An Express Node.js service (server/linkedinScraper.ts) crawls live LinkedIn vacancies to calculate hiring momentum, while serverless Firebase Cloud Functions v2 handle automated tech news ingestion and AI enrichment.
4. Database & Auth: User profiles, skill provenance tags, and generated 3D Transit Passes synchronize reactively with Google Cloud Firestore (NoSQL), authenticated securely through Firebase Auth.
5. Cloud Hosting: The complete full-stack prototype is containerized, deployed, and served live on Replit with SSL/HTTPS.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React 19.3	Core frontend UI rendering and reactive state	MIT

Name & Version	Purpose in Project	License
TypeScript 6.0	Static typing, interface contracts, and safety	Apache 2.0
Vite 8.3	Build tooling and development server	MIT
Tailwind CSS 4.3	Utility-first CSS styling and responsive layouts	MIT
Firebase SDK 10+	Cloud Firestore database and authentication client	Apache 2.0
Lucide React 1.6	Modern iconography and UI visual elements	ISC
Tesseract.js 7.0	Client-side Optical Character Recognition (OCR) for scanned CV images	Apache 2.0
pdfjs-dist 6.4	Client-side PDF text extraction and binary parsing	Apache 2.0
Mammoth 1.13	Client-side DOCX Word document reading and extraction	BSD-2-Clause

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API	Used to generate personalized career transition advice in the AI chat, predict weekly hiring shifts, and analyze the technical significance of incoming tech news articles.	Free Tier
LinkedIn Public Feed / Scraper	Aggregates real-time hiring vacancy counts, emerging skill deltas, and tech demand indices	Free
Tech News RSS Feeds (GitHub, Google, AWS)	Daily industry software and AI updates ingested into Catchup Hub	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Full-stack application hosting, live web server deployment, and dev environment	SDC Sponsor Credits
Google Cloud Firestore	NoSQL document database for profiles, session states, and career taxonomy	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 2.0 Flash / 2.5 Flash	Google / Google AI Studio	Pre-trained	Google Gemini API Terms of Service
Tesseract OCR (eng.traineddata)	Tesseract / Google Open Source	Pre-trained	Apache License 2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Real-Time Tech Job Postings Dataset	Aggregated live from public LinkedIn job postings	Public	Public Information	N
Tech News Articles Feed	Public RSS feeds from GitHub Blog, Google AI Blog, AWS Architecture, TechCrunch	Public	Public	N
JobTransit CS Career Metro Taxonomy	Curated internally based on MDEC, HRD Corp, and ACM/IEEE CS curricula	Self-collected	Apache License 2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✔ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Google Antigravity	Code assistance in debugging, unit test scaffolding, CSS styling	Moderate
ChatGPT	Project ideation, brainstorming	Minor
Replit	Hosting Deployment	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI tools were used for initial concept ideation, brainstorming problem definitions, and accelerating coding syntax. Our team critically evaluated each suggestion, discarded impractical concepts, and adapted the outputs to fit our project requirements. We reviewed all generated code for logic and security, debugged errors, verified the type definitions, and tested the prototype extensively to ensure full functionality and reliability.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✔Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Cloud Hosting Credits	Used to deploy and run the live production prototype web app and collaborative dev workspace.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Yes, because we need to host our interactive prototype publicly with zero setup friction. While the React/Node application is fully portable and can run on any standard container or cloud VM, Replit's integrated environment significantly streamlined our continuous integration and live demonstration pipeline for the hackathon.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

JobTransit adheres to privacy-by-design principles:

1. In-Browser Ingestion: Candidate resumes (PDF, DOCX, scans) are parsed entirely in-memory within the user's browser using PDF.js and Tesseract.js. Raw resume files are never stored on persistent server disks or cloud storage buckets.

2. Data Minimization: Only user-reviewed structured data (name, email, extracted skill tags, and transit milestones) is synchronized to Google Cloud Firestore, secured by Firebase Authentication and Firestore security rules.

3. Ephemeral AI Processing: Text sent to the Google Gemini API is transmitted via HTTPS strictly for real-time structured entity extraction and dynamic question generation. Under Google Gemini API non-enterprise terms, API data is not used to train base models.

4. Candidate Control: Users can clear browser session states or delete their Firestore profile data at any time.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

2. Rate Limits & Single-Provider Dependency: Relying on the Google Gemini API risks latency or quotas. Mitigation: Client-side dictionary algorithms (skills-catalog.ts) serve as an instant offline fallback if the API is throttled.
3. Web Scraping Volatility: Public job search layouts may change dynamically. Mitigation: Built an in-memory caching layer with automated fallback to verified historical demand baselines.
4. Domain Scope: To guarantee high taxonomy depth within the hackathon timeframe, pathways are currently focused specifically on Computer Science roles.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✔ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✔ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✔ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✔ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✔ Any use of official SDC sponsor resources has been declared in Section 5.
- ✔ The prototype described here is accessible to judges through the links provided in Section 1.
- ✔ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25 October 2026